

Implement A* Algorithm

Aim :

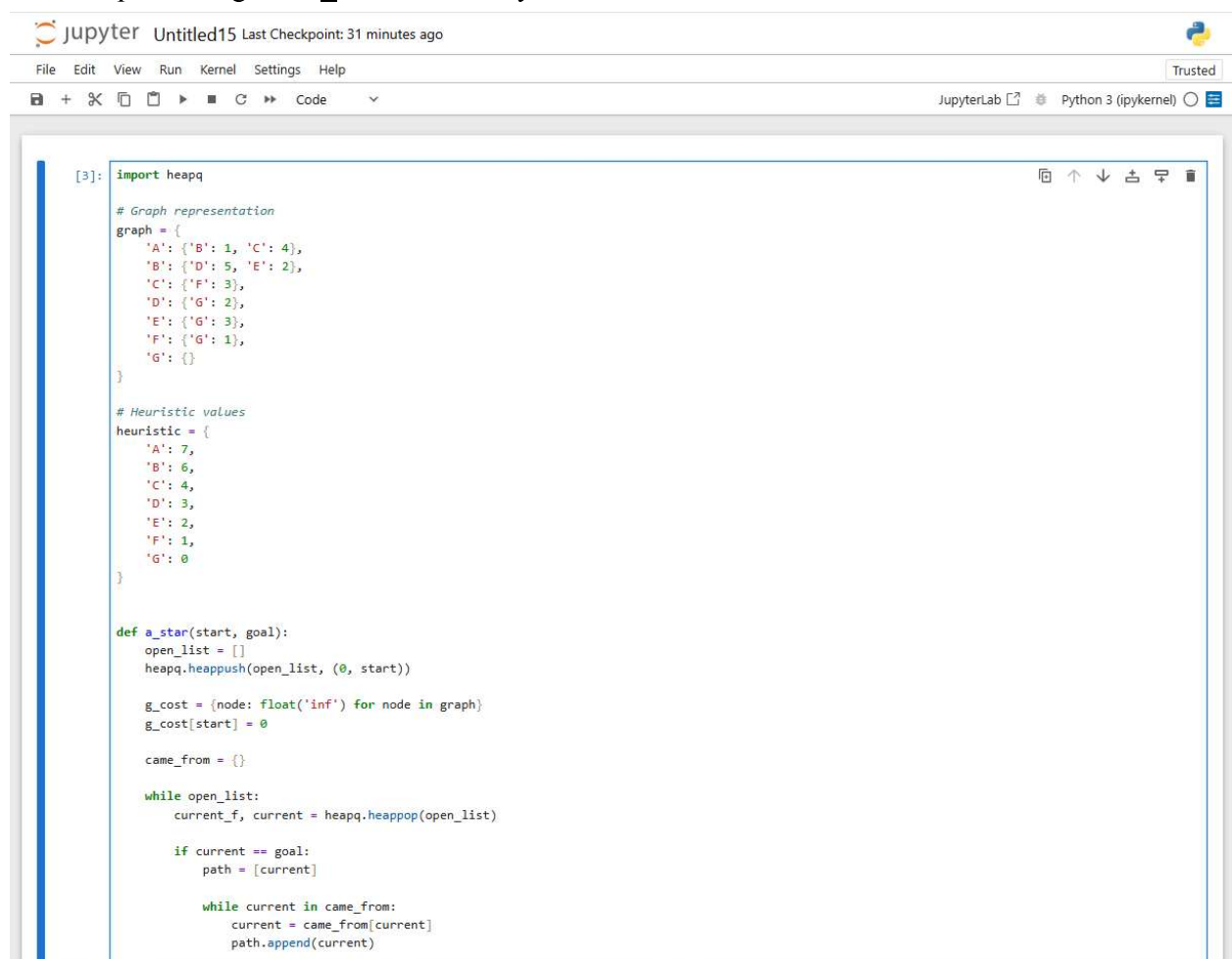
To Find the shortest path from a source to a destination using the A* algorithm with heuristics.

Explanation:

A* is a graph search algorithm that finds the shortest path using both actual cost and estimated cost (heuristic).

Procedure:

1. Initialize a priority queue with the start node.
2. Maintain $g(n)$ (path cost) and $h(n)$ (heuristic).
3. Expand the lowest $f(n) = g(n) + h(n)$ node.
4. Track path using `came_from` dictionary.



```
[3]: import heapq

# Graph representation
graph = {
    'A': {'B': 1, 'C': 4},
    'B': {'D': 5, 'E': 2},
    'C': {'F': 3},
    'D': {'G': 2},
    'E': {'G': 3},
    'F': {'G': 1},
    'G': {}
}

# Heuristic values
heuristic = {
    'A': 7,
    'B': 6,
    'C': 4,
    'D': 3,
    'E': 2,
    'F': 1,
    'G': 0
}

def a_star(start, goal):
    open_list = []
    heapq.heappush(open_list, (0, start))

    g_cost = {node: float('inf') for node in graph}
    g_cost[start] = 0

    came_from = {}

    while open_list:
        current_f, current = heapq.heappop(open_list)

        if current == goal:
            path = [current]

            while current in came_from:
                current = came_from[current]
            path.append(current)
```

```

        path.reverse()
        return path, g_cost[goal]

    for neighbor, cost in graph[current].items():
        tentative_g = g_cost[current] + cost

        if tentative_g < g_cost[neighbor]:
            came_from[neighbor] = current
            g_cost[neighbor] = tentative_g

        f_cost = tentative_g + heuristic[neighbor]
        heapq.heappush(open_list, (f_cost, neighbor))

    return None

# Driver code
start_node = 'A'
goal_node = 'G'

result = a_star(start_node, goal_node)

if result:
    path, cost = result
    print("Shortest Path:", path)
    print("Total Cost:", cost)
else:
    print("Path not found")

Shortest Path: ['A', 'B', 'E', 'G']
Total Cost: 6

```

[]:

Final Result:

The A* Search Algorithm was implemented successfully, and the optimal shortest path was obtained.